

Customer Churn EDA Report

Telco Customer Churn Dataset

Submitted by

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1.Introduction

Customer churn refers to the phenomenon where customers stop doing business with a company. In the telecom industry churn is a critical metric because acquiring a new customer costs significantly more than retaining an existing one.

Understanding why customers leave helps businesses take proactive steps to retain them. This report presents an exploratory data analysis of a telecom customer churn dataset to identify key patterns and factors that contribute to churn.

2. Dataset Details

Dataset Name	: Telco Customer Churn
Source	: Kaggle
Total Rows	: 7043
Total Columns	: 21
Rows after Cleaning	: 7032
Columns after Cleaning	: 20
Target Column	: Churn (Yes = churned, No = stayed)

Numerical Columns:

tenure, MonthlyCharges, TotalCharges, SeniorCitizen

Categorical Columns:

gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, churn

3. Data Cleaning Summary

The following cleaning steps were performed on the dataset:

1. Missing Values

TotalCharges column had 11 missing values. These 11 rows were dropped as they represent less than 0.2% of the total data. All other columns had zero missing values.

2. Duplicate Records

Checked for duplicate rows across all columns. No duplicate records were found in the dataset.

3. Unnecessary Columns

customerID column was dropped as it is just a unique identifier with no analytical value for churn analysis.

4. Data Type Corrections

All columns were verified for correct data types. No major data type issues were found.

5. Categorical Values Check

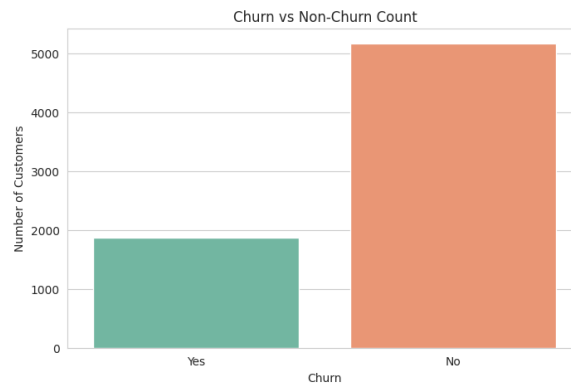
All categorical columns were checked for inconsistencies. No spelling mistakes or formatting issues were found. "No internet service" and "No phone service" were kept as separate categories as they represent distinct customer segments.

6. Outlier Check

IQR method was applied on tenure, MonthlyCharges and TotalCharges. No outliers were detected in any numerical column. All values are within normal bounds.

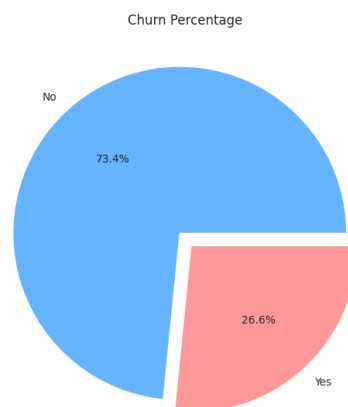
4. Exploratory Data Analysis

4.1 Bar Plot (Univariate)



A bar plot was plotted to show the count of churned vs non-churned customers. The chart clearly showed that non-churned customers are significantly higher than churned customers confirming the dataset is imbalanced.

4.2 Churn Percentage Pie Chart (Univariate)



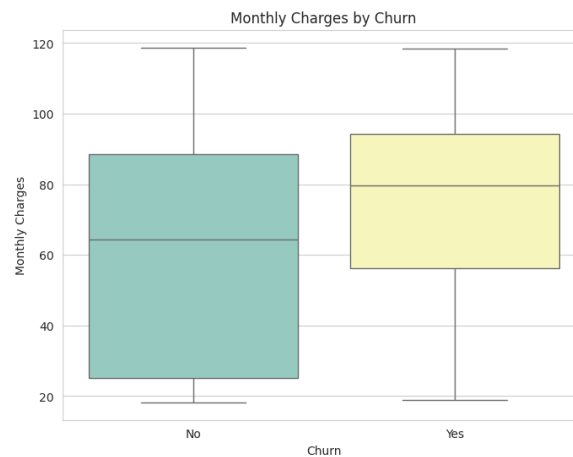
The pie chart revealed that 26.6% of customers churned while 73.4% stayed. This is a significantly high churn rate indicating a serious customer retention problem.

4.3 Tenure Distribution Histogram (Univariate)



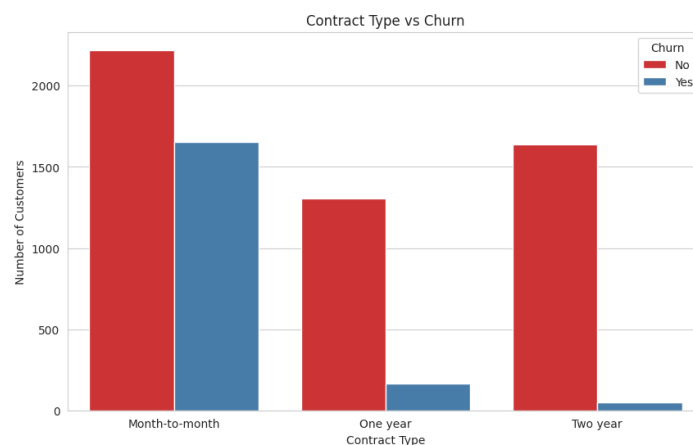
The histogram showed a U-shaped distribution. A large spike was observed at 0-2 months indicating many new customers. Another spike at 70+ months indicates a group of very loyal long term customers. The middle range was relatively stable.

4.4 Monthly Charges by Churn Boxplot (Bivariate)



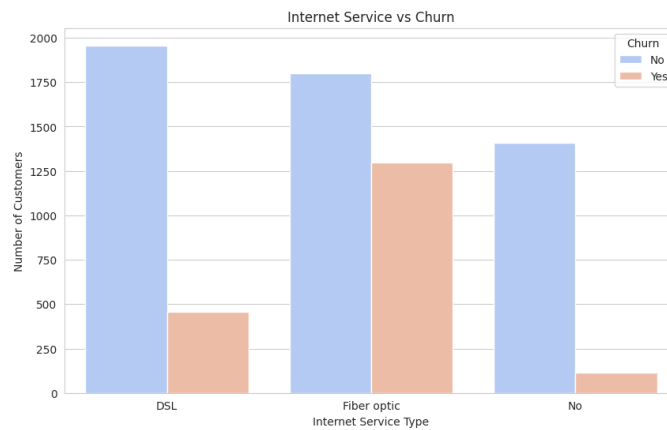
The boxplot showed that churned customers have higher median monthly charges (~80) compared to non-churned customers (~65). This suggests customers with higher bills are more likely to leave. No outliers were detected in either group.

4.5 Contract Type vs Churn (Bivariate)



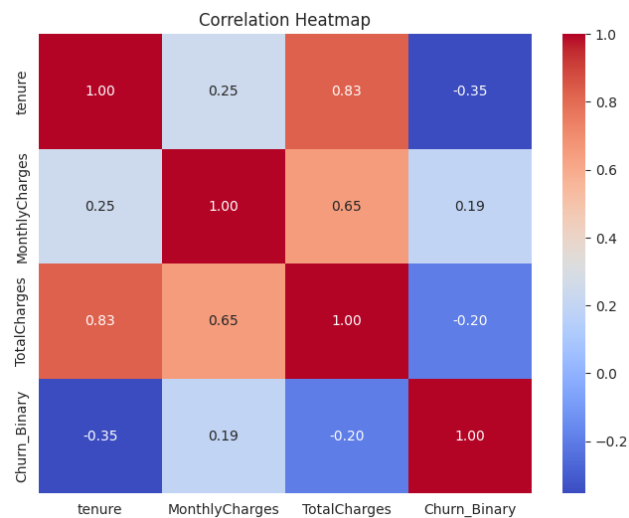
Month-to-month contract customers showed the highest churn (~1650). One year contract customers showed very little churn (~175) and two year contract customers almost never churned (~60). Longer the contract lower the churn — this is one of the strongest patterns found in the analysis.

4.6 Internet Service vs Churn (Bivariate)



Fiber optic customers showed the highest churn (~1300) despite having similar total customers as DSL. DSL customers showed moderate churn (~450). Customers with no internet service barely churned (~125). This suggests fiber optic customers may be dissatisfied with pricing or service quality.

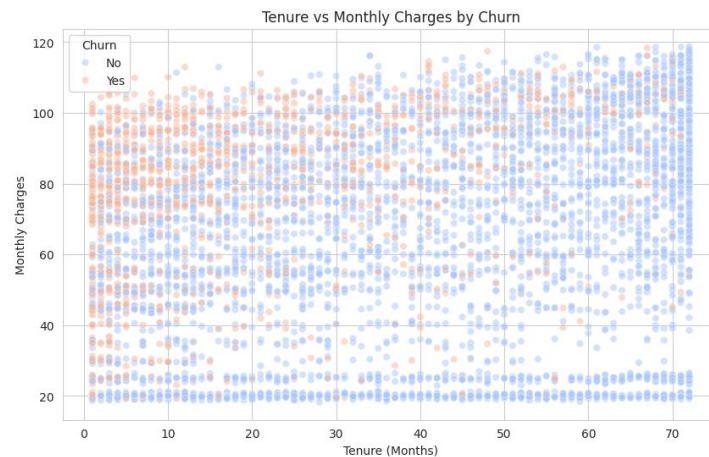
4.7 Correlation Heatmap



Tenure showed the strongest negative correlation with churn (-0.35) meaning longer staying customers are less likely to churn. Monthly charges showed a positive correlation (0.19) meaning higher charges lead to more churn. Tenure and TotalCharges were strongly correlated (0.83) which is expected as longer staying customers pay more overall.

4.8 Tenure vs Monthly Charges Scatter Plot (Bivariate)

Churned customers were heavily concentrated on the left side of the plot indicating low tenure. They were also visible at higher monthly charge levels. Non-churned customers dominated the right side confirming long term customers rarely churn.



4.9 Numerical Analysis

- Average tenure is 32 months with high variation (std dev 24.55).
- Monthly charges average at 64.80 ranging from 18.25 to 118.75.
- TotalCharges has a large gap between mean (2283) and median (1397) suggesting some very long term high paying customers.

4.10 Outlier Analysis

IQR method was applied on tenure, MonthlyCharges and TotalCharges. No outliers were detected in any column. All values fall within normal bounds confirming the dataset is clean and valid.

5. Key Findings

1. 26.6% of customers churned — roughly 1 in 4 customers left
2. Month-to-month contract customers churn the most
3. Fiber optic internet service customers have very high churn
4. New customers with low tenure (0-2 months) are most at risk
5. Customers with higher monthly charges tend to churn more
6. Long term customers (70+ months) are very loyal and rarely churn
7. Tenure is the strongest numerical predictor of churn (-0.35)
8. Contract type is the strongest categorical predictor of churn

6. Business Recommendations

1. Promote Long Term Contracts

Offer discounts and benefits to customers on month-to-month contracts to switch to one or two year contracts as this significantly reduces churn.

2. Early Retention Programs

Target new customers in their first 6 months with special offers and personalized engagement as they are most at risk of churning.

3. Investigate Fiber Optic Service

Fiber optic customers churn at very high rates. Investigate whether this is due to pricing or service quality issues and take corrective action immediately.

4. Review High Charge Customers

Offer flexible pricing plans or bundled discounts to customers with high monthly charges to prevent them from leaving.

5. Loyalty Rewards Program

Introduce rewards for customers who stay beyond 12 months to encourage long term retention and reduce churn.

7. Conclusion

This exploratory data analysis of the Telco Customer Churn dataset revealed several important patterns. The overall churn rate of 26.6% indicates a significant retention problem. The analysis identified contract type, tenure and monthly charges as the top factors affecting churn. Month-to-month contract customers and fiber optic users are the most at risk groups.

By implementing targeted retention strategies such as promoting long term contracts, early engagement programs and reviewing fiber optic pricing the company can significantly reduce its churn rate and improve overall customer satisfaction.